HR Analytics Dashboard using Python, SQL & Power BI

📌 Overview

This project focuses on analyzing HR employee data using Python, SQL, and Power BI to uncover workforce trends, employee attrition patterns, salary distribution, job satisfaction levels, and performance insights.

The objective is to transform raw HR data into meaningful business insights that help organizations improve employee retention and workforce management.

📊 Data Analysis using Python

Performed Exploratory Data Analysis (EDA) using Python to:

- Check data quality and consistency
- Analyze employee demographics
- Study employee attrition patterns
- Analyze department-wise employee distribution
- Examine salary distribution across job roles
- Evaluate job satisfaction and performance ratings
- Generate statistical insights and visualizations

🗄️ Data Analysis using SQL

Performed business analysis using SQL to:

- Calculate total monthly income by gender
- Analyze attrition across departments
- Compare average income by job role
- Identify education fields with high attrition
- Evaluate job satisfaction trends
- Find experienced high-performing employees
- Rank employees based on income
- Identify top-paid employees by department
- Detect high performers with below-average salaries
- Analyze tenure and attrition trends
- Compare promotion potential across genders
- Study the impact of overtime on income

📈 Power BI Dashboard

Created an interactive HR Analytics Dashboard to visualize workforce insights.

Dashboard Includes

- Employee Count
- Attrition Count & Attrition Rate
- Department Analysis
- Gender Analysis
- Salary Analysis
- Job Satisfaction Analysis
- Job Role Analysis
- Performance Analysis
- Overtime Analysis
- Workforce Distribution

💡 Key Insights

- Research & Development has the highest employee count.
- Attrition is concentrated within specific departments and job roles.
- Employees with lower compensation are more likely to leave.
- Job satisfaction varies across different roles.
- Overtime significantly impacts employee retention and income.
- Some high-performing employees receive below-average salaries.

🛠️ Technologies Used

- Python
- Pandas
- Matplotlib
- SQL
- Power BI